

CSS Layout and Border Properties Guide

1. CSS Border Style

Concept

The border-style property specifies the type of border displayed around an element.

Syntax

```
selector {  
    border-style: value;  
}
```

Common Values

- solid
- dashed
- dotted
- double
- groove
- ridge
- inset
- outset
- none

Example

```
<div class="box">Border Style Example</div>
```

```
.box {  
    border-style: dashed;  
    border-width: 3px;  
}
```

2. CSS Border Width

Concept

Defines the thickness of the border.

Syntax

border-width: value;

Example

```
.box {  
    border-style: solid;  
    border-width: 5px;  
}
```

3. CSS Border Color

Concept

Sets the color of the border.

Syntax

border-color: color;

Example

```
.box {  
    border-style: solid;  
    border-width: 3px;  
    border-color: red;  
}
```

4. CSS Border Sides

Concept

Apply borders to specific sides of an element.

Properties

border-top

border-right

border-bottom

border-left

Example

```
.box {  
    border-top: 3px solid red;  
    border-bottom: 3px solid blue;  
}
```

5. CSS Border Shorthand

Concept

Combine border width, style, and color into one property.

Syntax

border: width style color;

Example

```
.box {  
    border: 2px solid green;  
}
```

6. CSS Rounded Borders

Concept

The border-radius property creates rounded corners.

Syntax

border-radius: value;

Example

```
.box {  
    border: 2px solid black;  
    border-radius: 15px;  
}
```

Circle Example

```
.circle {  
    width: 100px;  
    height: 100px;  
    border-radius: 50%;  
    background: lightblue;  
}
```

CSS Margins

7. Margin

Concept

Margins create space outside an element's border.

Syntax

```
margin: value;
```

Example

```
.box {  
    margin: 20px;  
}
```

Individual Sides

```
.box {  
    margin-top: 10px;
```

```
margin-right: 20px;
margin-bottom: 30px;
margin-left: 40px;
}
```

Shorthand

```
.box {
  margin: 10px 20px 30px 40px;
}
```

8. Margin Collapse

Concept

Vertical margins between adjacent block elements collapse into a single margin.

Example

```
<div class="box1">Box 1</div>
```

```
<div class="box2">Box 2</div>
```

```
.box1 {
  margin-bottom: 30px;
}
```

```
.box2 {
  margin-top: 20px;
}
```

Actual space between elements:

30px (not 50px)

CSS Padding

9. Padding

Concept

Padding creates space between content and the border.

Syntax

```
padding: value;
```

Example

```
.box {  
    padding: 20px;  
    border: 1px solid black;  
}
```

Individual Sides

```
.box {  
    padding-top: 10px;  
    padding-right: 20px;  
    padding-bottom: 30px;  
    padding-left: 40px;  
}
```

Shorthand

```
.box {  
    padding: 10px 20px 30px 40px;  
}
```

10. box-sizing

Concept

Controls how element dimensions are calculated.

Values

- content-box (default)
- border-box

Example

```
.box {  
  width: 200px;  
  padding: 20px;  
  border: 5px solid black;  
  box-sizing: border-box;  
}
```

Benefit

The total width remains 200px including padding and border.

CSS Height and Width

11. Height / Width

Concept

Defines the dimensions of an element.

Example

```
.box {  
  width: 300px;  
  height: 150px;  
}
```

Responsive Example

```
.box {  
  width: 80%;  
  height: auto;
```

}

12. Min / Max Width

Concept

Sets minimum and maximum width limits.

Example

```
.box {  
    width: 80%;  
    min-width: 300px;  
    max-width: 800px;  
}
```

Use Case

Useful for responsive web design.

CSS Box Model

13. Box Model

Concept

Every HTML element is represented as a rectangular box.

Structure

```
+-----+  
|  Margin  |  
| +-----+ |  
| | Border | | | |
| | +-----+ | |  
| | | Padding | | |  
| | | +-----+ | | |
```


| | | Content | | |

| | | +-----+ | |

| | +-----+ | |

| +-----+ |

+-----+

Example

```
.box {  
  width: 200px;  
  padding: 20px;  
  border: 5px solid black;  
  margin: 30px;  
}
```

Total Width Calculation

Width + Left Padding + Right Padding
+ Left Border + Right Border

CSS Outline

14. Outline Style

Concept

Defines the appearance of an outline drawn outside the border.

Example

```
.box {  
  outline-style: dashed;  
}
```

15. Outline Width

Concept

Specifies the thickness of the outline.

Example

```
.box {  
    outline-style: solid;  
    outline-width: 4px;  
}
```

16. Outline Color

Concept

Specifies the color of the outline.

Example

```
.box {  
    outline-style: solid;  
    outline-color: blue;  
}
```

17. Outline Shorthand

Concept

Combines width, style, and color.

Syntax

outline: width style color;

Example

```
.box {  
    outline: 3px solid red;  
}
```

18. Outline Offset

Concept

Creates space between the border and the outline.

Example

```
.box {  
    border: 2px solid black;  
    outline: 3px solid red;  
    outline-offset: 10px;  
}
```

Complete Example

```
<!DOCTYPE html>  
  
<html>  
  
<head>  
  
<style>  
  
.card {  
    width: 300px;  
    padding: 20px;  
    margin: 30px auto;  
  
    border: 3px solid #3498db;  
    border-radius: 10px;  
  
    outline: 2px dashed red;  
    outline-offset: 5px;
```

```
    box-sizing: border-box;
}
</style>
</head>
<body>

<div class="card">
    <h2>CSS Box Model Example</h2>
    <p>This element demonstrates margin, border, padding, outline, and sizing.</p>
</div>

</body>
</html>
```